THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 105	Complete graph on n points: $\frac{n(n-1)}{2}$.
2	**(a)** 150	Find the new time first, then the angle: each hour mark is 30°.
3	**(d)** 5	Repeat the largest-minus-smallest step.
4	**(b)** 140	Total minus the largest.
5	**(a)** 1001	Divide 1000 by the number and round up.
6	**(d)** 37	Try triangular numbers.
7	**(b)** 1170	$360n + 90$.
8	**(c)** 24	List the sequence.
9	**(c)** 4	Divide by 3 (the symbols are in the same proportion).
10	**(a)** 19	Total numbers − primes − 1 (1 is neither).
11	**(b)** 800	Subtract the first 10 odd numbers, whose sum is 100.
12	**(d)** 180	Net turn.
13	**(a)** Both A and R are true and R is the correct explanation of A.	15 + 4 = 19, and R is the rule of the pattern.
14	**(a)** Both A and R are true and R is the correct explanation of A.	135 lies between 90 and 180, and R is exactly the definition.
15	**(a)** Both A and R are true and R is the correct explanation of A.	7, 5, 3, 0 in descending order gives 7530.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Complete graph idea: each pair plays once (1); $\frac{14\times13}{2} = 91$ (1).	1+1
17	Hour hand points to 4 on the dial; smaller angle $= 120^\circ$ (1); Hour hand turns $2 \times 30 = 60^\circ$ in 2 hours (1).	1+1
18	Largest minus smallest each time; $1234 \to 3087 \to 8352 \to 6174$ (1); Steps $= 3$ (1).	1+1
19	Milk $= 150 - 62 - 48$ (1); 40 people (1).	1+1
20	LCM of 8, 12, 18 (1); $= 72$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$a + (n-1)d = 365$ gives $n - 1 = (365 - 14) \div 9$ (1); $n = 40$ (1); Term before it $= 356$ (1).	1+1+1
22	Hour hand points to 1 on the dial; smaller angle $= 30^\circ$ (1); Larger angle $= 360 - 30 = 330^\circ$ (1); Reflex angle: the larger angle 330° (1).	1+1+1
23	Number 1551 $= 1551$; $1551 \div 11 = 141$ exactly (1); Digit sum $= 12$ (1); Palindrome reads the same backwards (1).	1+1+1
24	Week 1 total $= 61$; Week 2 total $= 60$ (1); Larger: Week 1 (1); Difference $= 1$ (1).	1+1+1
	OR: LCM of 48, 72, 108 is 432 seconds (1); That is 7 minutes 12 seconds (1); Time: 8:07:12 am (1).	1+1+1
25	First 20 odd numbers sum to 400 (1); First 25 odd numbers sum to 625 (1); Required sum $= 225$ (1).	1+1+1
	OR: Minutes $= 90 \div 6 = 15$ (1); Right angles $= 90 \div 90 = 1$ (1); Time shown: 2:15 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) 13 sticks (1). (ii) 3 sticks (1). (iii) $4 + 9 \times 3 = 31$ sticks (1). (iv) $40 - 4 = 36$ and $36 \div 3 = 12$ more squares, so $1 + 12 = 13$ squares (2).	1+1+1+2

27	Number 12321 (1); Digit sum 9 (1); Divisible by 9 (1); Smallest number to add: 0 (1); Check: new digit sum is a multiple of 9 (1).	1+1+1+1+1
	OR: A $= 36$, B $= 72$, C $= 108$ (1); D $= 36 + 72 - 108$ (1); D $= 0$ (1); Height of D $= 0$ cm (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 39	$20^2 - 19^2$ is the next odd number.
2	**(c)** 30	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
3	**(b)** 165	Add the numbers.
4	**(b)** 9	Add the children, then divide by the value of one symbol.
5	**(b)** 24	HCF of three numbers.
6	**(a)** 11	Divide the total increase by the number of steps, which is one less than the term number.
7	**(d)** 25	The minute hand turns 6° per minute.
8	**(b)** 24	Subtract.
9	**(a)** 37	Subtract.
10	**(b)** 5	Digit sum must be a multiple of 9.
11	**(a)** 88	Add the triangular number.
12	**(b)** 4	Each line gives one straight angle.
13	**(a)** Both A and R are true and R is the correct explanation of A.	3.5 x 10 = 35, and R says what the half symbol means.
14	**(a)** Both A and R are true and R is the correct explanation of A.	A prime has exactly two factors and a composite has more than two; 1 has just one factor.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	1+3+5+7+9 = 25, and 25 = 5 x 5 is a square, but R does not explain why the sum is 25.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$n \times n = 961$ so $n = 31$ (1); 31th odd number $= 2 \times 31 - 1 = 61$ (1).	1+1
17	Total $= 360 \times 2 + 90 = 810^\circ$ (1); Right angles $= (810) \div 90 = 9$ (1).	1+1
18	Apply the rule (1); Terms: 15, maximum: 52 (1).	1+1
19	First bar $= 180$; Second bar height $= 6$ cm (1); Value $= 120$ (1).	1+1
20	The numbers include a multiple of 3 (1); Hence the product is divisible by 3 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Build the row; Row: 2, 3, 5, 8, 13, 21, 34, 55 (1); 8th number $= 55$ (1); Sum $= 141$ (1).	1+1+1
22	Pens $= 36$ (1); Notebooks $= 72$ (1); Total $= 108$ (1).	1+1+1
23	Subtract the remainder: 72 and 96 (1); HCF of 72 and 96 is 24 (1); The number is 24 (it must exceed 4) (1).	1+1+1
24	Row 6 has $1 + 2 + \cdots + 6 = 21$ cubes (1); First rows: 1+3+6+10 (1); Total in first 4 rows $= 20$ (1).	1+1+1
	OR: Slices $= 2 \times 2 = 4$ (1); Angle $= 360 \div 4 = 90^\circ$ (1); Type: right (1).	1+1+1
25	Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 9752, smallest 2579 (1); Difference 7173 (1).	1+1+1
	OR: Frequencies: 1: 1; 2: 1; 3: 1; 4: 1; 5: 4; 6: 1; 7: 2; 8: 1 (1); Values greater than 5: 1 + 2 + 1 $= 4$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $20 \times 6 = 120^\circ$ (1). (ii) 30° (1). (iii) At 4 o'clock the angle is $4 \times 30 = 120^\circ$, an obtuse angle (1). (iv) At 7 o'clock one angle is 210° and the smaller angle is $360^\circ - 210^\circ = 150^\circ$ (1); the 210° angle is a reflex angle (1).	1+1+1+2

27	Net $= 180 - 45 + 90 = 225^\circ$ (1); Direction 225° clockwise from North (1); To face North again: turn clockwise 135° more (1); Check: $225 + 135 = 360^\circ$ (2).	1+1+1+2
	OR: Reverse-and-add repeatedly (1); Sums: 143, 484 (1); Steps $= 2$ (1); Palindrome 484 has 3 digits (odd) (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 2025	Square of the triangular number.
2	**(c)** 72	$360^\circ \div 5$.
3	**(c)** 968	Reverse and add.
4	**(b)** 8	Integer division.
5	**(c)** 41	List the primes after 10.
6	**(b)** 41	Each triangular number adds the next counting number.
7	**(b)** 150	Multiply the angle between neighbours by the number of gaps.
8	**(c)** 3	Continue the process.
9	**(c)** 7.5	Divide by the scale.
10	**(b)** 97	Check numbers just below.
11	**(b)** 153	Complete graph on n points: $\frac{n(n-1)}{2}$.
12	**(c)** 90	Find the new time first, then the angle: each hour mark is 30°.
13	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but the value of a right angle does not explain the value of a straight angle.
14	**(a)** Both A and R are true and R is the correct explanation of A.	7 is odd: 3 x 7 + 1 = 22.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true: equal widths keep the graph fair, heights show the data, but R does not explain A.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Sum of first 24 odd numbers $= 576$ and first 9 odd numbers $= 81$ (1); Required sum $= 495$ (1).	1+1
17	Each pair of rays makes one angle (1); $\frac{8\times7}{2} = 28$ (1).	1+1
18	Digit sum $= 12 \times 5 = 60$ (1); Divisible by 3: yes; divisible by 9: no (1).	1+1
19	Total $= 2+3+1+4+0+5 = 15$ (1); Matches with more than 2 goals: 3 (1).	1+1
20	$84 = 2^2 \times 3 \times 7$ (1); Needed multiplier: 15 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	LCM $= 24$ (1); Required number $= LCM + 5 = 29$ (1); Check each remainder (1).	1+1+1
22	$a + (n-1)d = 331$ gives $n - 1 = (331 - 23) \div 7$ (1); $n = 45$ (1); Term before it $= 324$ (1).	1+1+1
23	Hour hand points to 5 on the dial; smaller angle $= 150^\circ$ (1); Larger angle $= 360 - 150 = 210^\circ$ (1); Reflex angle: the larger angle 210° (1).	1+1+1
24	Number 2882 $= 2882$; $2882 \div 11 = 262$ exactly (1); Digit sum $= 20$ (1); Palindrome reads the same backwards (1).	1+1+1
	OR: Week 1 total $= 64$; Week 2 total $= 62$ (1); Larger: Week 1 (1); Difference $= 2$ (1).	1+1+1
25	LCM of 30, 45, 60 is 180 seconds (1); That is 3 minutes 0 seconds (1); Time: 8:03:00 am (1).	1+1+1
	OR: First 25 odd numbers sum to 625 (1); First 31 odd numbers sum to 961 (1); Required sum $= 336$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) 8530 (1). (ii) 3058 (1). (iii) $8530 - 3058 = 5472$ (1). (iv) An odd number must end in 5 or 3. The smallest start is 3 then 0, giving 3058 (even) or 3085; 3085 ends in 5 and is the smallest odd number (2).	1+1+1+2

27	Figure 15: dots $= 5 + 14 \times 9 = 131$ (1); For 176 dots: steps from figure $1 = (176 - 5) \div 9 = 19$ (1); Figure number $= 20$ (1); Check: substitute back to get the same number of dots (2).	1+1+1+2
	OR: (a) Hour hand: 4 hour marks of $30^\circ = 120^\circ$ (1); (b) Minute hand has completed 4 full turns $= 1440^\circ$ and stands at 12 (1); (c) Smaller angle $= 120^\circ$ (1); (d) obtuse angle (2).	1+1+1+2

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